

The Burden Shift Is Not Enough: Rebuttal Technology and Deterrence in Artificial Intelligence Liability

Ignacio Adrián Lerer

Working paper. First of three papers derived from Lerer (2026a).

Abstract

Liability rules for artificial intelligence are ordinarily compared on the margin that classical accident law treats as decisive: whether the injurer's care should be assessed against a due-care standard, as under negligence, or bypassed entirely, as under strict liability. I argue that for harm mediated by artificial intelligence the binding margin is evidentiary rather than behavioural, and I model it accordingly. The model is a two-stage game in which a developer chooses an unobservable safety investment and harm then realises with a probability decreasing in that investment. Verification asymmetry enters through the probability that the party carrying the burden of proof can discharge it. Three regimes are compared: strict liability with the claimant's burden, negligence with the claimant's burden, and strict liability with a rebuttable presumption of defect. Five results follow. A claimant's burden dilutes deterrence in proportion to the probability of proof, so care decays continuously toward zero. Negligence collapses discontinuously below a threshold, and because proving breach requires establishing both causation and the counterfactual standard, negligence is more demanding of verification than strict liability. A presumption whose rebuttal probability does not respond to care is algebraically equivalent to claimant-burden strict liability with a relabelled parameter, so the burden shift alone buys no deterrence; this identifies an assumption left implicit in the existing literature on presumptions of negligence. When rebuttal is care-responsive, care purchases exculpatory evidence and deterrence is largely restored. Whether the regime under-deters or over-deters turns on a comparison of two elasticities, which yields a design rule for safe harbours. Extensions to bilateral care, litigation costs, judgment-proof developers, endogenous rebuttal, and activity levels are examined. The results suggest that the presumption introduced by the revised European product liability regime will deter only if regulatory compliance certification is not accepted as sufficient rebuttal.

Keywords: artificial intelligence liability; burden of proof; rebuttable presumption; evidentiary asymmetry; deterrence; strict liability; negligence; safe harbour design; Product Liability Directive 2024/2853; law and economics; accident law; extended phenotype theory

1. Introduction

The question of who should bear the cost of harm caused by an artificial intelligence system is usually posed as a choice between two familiar rules. Under negligence, the developer pays if its care fell below a standard the court sets. Under strict liability, the developer pays whenever its product caused the harm, whatever care it took. Shavell (1987) supplies the canonical apparatus for choosing between them, and the choice turns on which party controls the margin that matters: care, activity level, or both.

That apparatus assumes something that artificial intelligence removes. In the accident settings for which it was built, the conduct the rule prices is discoverable. A driver's speed, a factory's emissions, and a manufacturer's quality-control records can be reconstructed by inspection, discovery, and expert testimony, at a cost that is high but bounded, and that falls on both parties in roughly comparable measure. When the injuring agent is a trained model whose failure modes are probabilistic, rarely reproducible on demand, and often not fully characterised even by the party that built it, that assumption fails. It fails asymmetrically: the developer retains privileged access to the training data, the evaluation protocol, and the incident record, while the affected party confronts an opacity that spending more on litigation does not dissolve.

Lerer (2026a) argues that this asymmetry is structural rather than transactional, meaning that it does not shrink as bargaining costs fall, and that recognising it is a precondition for any serious answer to the liability question. That argument was made conceptually. Here I formalise it, and the formalisation changes the answer in a way the conceptual treatment did not anticipate.

The change comes from locating the asymmetry correctly. It does not operate on the care margin, where accident theory places the comparison between negligence and strict liability. It operates on the evidentiary margin: on the probability that whichever party bears the burden of proof can discharge it. Once the model is built on that margin, the ranking of rules shifts, and a device that classical theory treats as a procedural detail, namely the allocation of the burden of proof, becomes the principal instrument.

The sharper finding is that this instrument is worthless when used alone. A presumption that a developer can rebut by producing evidence unrelated to the care it actually took reproduces, exactly, the regime it was meant to replace. What restores deterrence is not the direction in which the burden runs but the shape of the rebuttal standard: whether escaping liability requires the producer to show the very safety work the rule exists to encourage. This distinction has a policy edge. The revised European product liability regime introduces exactly such a presumption, and whether it deters at all will be settled not by the instrument itself but by what national transposition accepts as sufficient rebuttal.

The result also speaks to an open question in the economics of proof burdens. Hay and Spier (1997) analysed the allocation of burdens as a matter of minimising the costs of adjudication, and a line of work following them treated the allocation as broadly neutral with respect to primary conduct. Guerra, Luppi and Parisi (2022) challenged that neutrality, arguing that presuming negligence strengthens injurer care where claimants face evidentiary difficulty. Their result rests on a step they leave implicit: that only a party who invested in care can rebut the presumption. The model below makes that step an explicit parameter and shows that the entire deterrent effect depends on it. Where rebuttal is available on grounds unconnected to care, neutrality returns, and returns exactly rather than approximately.

2. Related literature and positioning

Epistemological positioning. This paper belongs to a research programme (Lakatos, 1978) that applies Generalized Darwinism and Extended Phenotype Theory (Dawkins, 1982) to legal and institutional systems. Popper (1959) supplies the demarcation criterion: the propositions formulated here are

falsable, and the conditions under which each would fail are stated with the proposition. Lakatos provides the architecture: the hard core of the programme, that legal institutions evolve through variation, selection, and retention, is protected by an auxiliary belt of specific models and metrics that are individually revisable without abandoning the core commitment. Hull (1988) supplies the account of how such revision proceeds through the competitive dynamics of a scholarly community rather than by individual falsification alone. The model advanced here belongs to the auxiliary belt: it can be rejected on its own terms without disturbing the programme's core.

2.1. Entitlements and the care margin

Coase (1960) established that when transaction costs are positive the initial assignment of entitlements determines the efficiency of the outcome, and Calabresi and Melamed (1972) turned that insight into a choice between protecting an entitlement by a property rule, which requires consent, or by a liability rule, which permits taking upon payment of judicially determined compensation. Shavell (1987) took the entitlement as given and compared negligence with strict liability on their incentive properties, showing that where only the injurer's care matters both rules can induce optimal care, while they differ once activity levels or victim precautions enter.

All three treat verification as costly but symmetric in the relevant sense: neither party enjoys a structural advantage in establishing what happened, and where courts cannot observe care at all the standard response is to prefer strict liability, which requires no such observation. That response is the one artificial intelligence complicates, because strict liability still requires the claimant to establish defect and causation, and it is there that the asymmetry bites.

2.2. The economics of proof burdens

A separate literature treats the burden of proof as an economic instrument in its own right. Hay and Spier (1997) analysed which party should bear the burden, framing the allocation largely in terms of the costs of producing evidence and of adjudicative error. Demougin and Fluet (2006) examined the preponderance standard and its incentive properties, and Kaplow (2011) analysed the optimal burden of proof directly in terms of the deterrence and chilling effects that different evidentiary thresholds produce. Taken together, that line of work leaves little doubt that evidentiary rules are not incentive-neutral.

Guerra, Luppi and Parisi (2022) brought the two literatures together, asking whether presumptions of negligence induce optimal precautions. Working with a binary care choice and symmetric evidentiary error, they argue that presuming negligence strengthens injurer incentives where claimants face proof difficulties. The model developed below differs in three respects. Care is continuous rather than binary, so the comparison is between levels rather than between compliance and its absence. The rebuttal probability is an explicit function of care rather than a fixed error rate. And the analysis identifies the condition under which their conclusion holds and the condition under which it fails exactly, which is the paper's principal contribution to that debate.

2.3. Institutional persistence

The second frame is the account of institutional persistence developed in Lerer (2025a, 2026a), under which a legal rule survives according to the product of its probabilities of transmission, compliance, and enforcement, rather than according to its static efficiency. A rule can be efficient on Shavell's criteria and still decay into what that literature calls a zombie-law equilibrium: formally in force, widely taught, and practically inert. The model here identifies one mechanism by which an artificial intelligence liability rule can reach that state, namely an evidentiary burden that the party carrying it cannot discharge, and it identifies the design feature that prevents it.

3. The model

3.1. Primitives and timing

A developer D deploys a system. In the first stage D chooses a safety investment x greater than or equal to zero at cost $c(x)$, with c increasing, strictly convex, and $c(0) = 0$. The investment is not observed by the affected party and is not directly verifiable by a court. In the second stage harm of magnitude H occurs with probability $p(x)$, where p is decreasing and strictly convex. The affected party V is treated as a passive victim in the baseline; Section 6.1 relaxes this.

The following functional forms are adopted as modelling assumptions rather than as findings, and Section 7 reports the extent to which the results survive changes to them:

$$c(x) = kx^2 / 2, \quad p(x) = p_0 \exp(-ax), \quad H > 0$$

3.2. The social optimum

A planner who observed x would minimise the sum of care cost and expected harm, so that the socially optimal care x^* satisfies

$$c'(x^*) = -p'(x^*) H$$

Under full verification, strict liability implements this exactly, because the developer internalises the entire expected harm and therefore faces the planner's problem. The benchmark is standard and is reproduced here only to fix notation.

3.3. The evidentiary margin

The departure from accident theory is in the location of the informational friction. I do not assume that the court cannot observe care and therefore cannot administer a negligence standard, which is the familiar treatment and which points directly to strict liability as the remedy. I assume instead that establishing the elements of the claim is probabilistic, and that the probability depends on which party must establish them.

Let σ in the interval from zero to one denote the probability that the affected party, bearing the burden, succeeds in proving defect and causation. Under the structural asymmetry described in Lerer (2026a), σ is low and bounded away from one however much V spends, because the evidence required is generated at

the point of design and is held by D. The parameter σ accordingly operationalises, on the evidentiary margin, the component of interaction cost that Lerer (2025b) treats as structural, meaning invariant to how often the interaction repeats.

Let σ_N denote the corresponding probability under a negligence rule, where V must prove not only defect and causation but also that care fell below the standard. The event in which the claimant establishes breach is contained in the event in which the claimant establishes defect and causation, since the former requires the latter together with the counterfactual standard. Hence σ_N is no greater than σ , and this ordering is a property of the evidentiary structure rather than an assumption about parameter values.

3.4. Rebuttal technology

Under a presumption regime the burden runs the other way. Defect and causation are presumed, and D escapes liability by rebutting. Let $p(x)$ denote the probability of successful rebuttal, taken to be differentiable and valued in the interval from zero to one.

The central modelling choice of this paper is to allow p to depend on x . The justification is evidentiary rather than technical: a developer that actually invested in safety holds documentation of having done so, namely evaluation records against the failure modes at issue, red-team results, and incident logs, while a developer that did not invest holds no such record and must rebut on other grounds. Whether the legal standard for rebuttal permits those other grounds is precisely the design question, and the parameter p' measures the answer. A rebuttal standard that accepts formal certification unconnected to the deployed system's risk profile yields p' near zero. A standard that requires system-specific evidence of safety work yields p' strictly positive. Section 6.4 derives p from a documentation technology rather than postulating it.

4. Equilibrium under the three regimes

Proofs of all propositions are collected in Appendix A.

4.1. Strict liability with the claimant's burden

Expected liability is $\sigma p(x) H$, so D solves the problem of minimising $c(x) + \sigma p(x) H$, giving the first-order condition

$$c'(x_C) = -\sigma p'(x_C) H$$

Proposition 1 (dilution). *The developer's objective is strictly convex, so x_C is unique. For σ less than one, x_C is strictly less than x^* , is strictly increasing in σ , and tends to zero as σ tends to zero. Deterrence decays continuously in the probability of proof.*

The mechanism is that a claimant's burden operates as a uniform discount on expected liability. It does not distinguish between a developer that invested and one that did not, because σ is exogenous to x . Both

face the same probability of escaping liability, and the rule therefore prices care at a fraction of its social value.

4.2. Negligence with the claimant's burden

Let $\bar{x}$ denote the due-care standard, set at the socially optimal level. D either meets it, incurring $c(\bar{x})$ and no liability, or falls short and faces expected liability $\sigma_N p(x) H$. Compliance is chosen when $c(\bar{x})$ is no greater than the minimised cost of shirking.

Proposition 2 (threshold collapse). *There exists a unique threshold $\hat{\sigma}$ in the open interval from zero to one such that D meets the standard when σ_N is at least $\hat{\sigma}$ and abandons it entirely when σ_N is below $\hat{\sigma}$, in which case equilibrium care coincides with that of Proposition 1 evaluated at σ_N . Since σ_N is no greater than σ , negligence is more demanding of verification than strict liability.*

The contrast with Proposition 1 is in the shape of the failure rather than in its existence. Strict liability degrades gracefully: as proof becomes harder, care falls smoothly. Negligence does not degrade at all until it fails completely. Above the threshold it implements the standard exactly; below it, the standard is ignored and the regime delivers the same outcome as diluted strict liability. A regulator calibrating a negligence rule therefore faces a discontinuity precisely in the region where artificial intelligence cases are likely to sit, and the discontinuity is not observable from the rule's performance in easier cases.

4.3. Strict liability with a rebuttable presumption

Expected liability is $p(x)(1 - p(x))H$, so D minimises $c(x) + p(x)(1 - p(x))H$, giving

$$c'(x_P) = -p'(x_P)(1 - p(x_P))H + p(x_P)p'(x_P)H$$

Two channels appear. The first is the familiar harm-reduction channel, attenuated by the probability of rebuttal. The second has no counterpart in accident theory: investing in safety raises the probability of exculpation, so care purchases evidence as well as safety.

Proposition 3 (irrelevance of the burden shift alone). *If the rebuttal probability is care-insensitive, so that $p(x)$ equals a constant $\bar{p}$ and p' equals zero, the first-order condition reduces to $c'(x) = -(1 - \bar{p})p'(x)H$, which is the condition of Proposition 1 with σ replaced by $1 - \bar{p}$. The two regimes are then observationally equivalent, and the burden shift changes nothing except the label of the parameter.*

This is an algebraic identity rather than an approximation, and it carries the policy weight of the paper. A legislature that shifts the burden of proof onto the developer while permitting rebuttal by means unconnected to the care actually taken has redistributed litigation risk without altering the incentive to take care. It also locates the implicit step in Guerra, Luppi and Parisi (2022): their deterrent effect is generated by the assumption that only an investing party can rebut, which in the present notation is the assumption that p' exceeds zero.

Proposition 4 (care-responsive rebuttal). *If p is differentiable and strictly increasing and the developer's objective is strictly convex on the relevant range, the term $p(x) - p'(x)H$ is strictly positive and equilibrium care exceeds the level obtaining under a care-insensitive rebuttal fixed at the same probability.*

Proposition 5 (elasticity criterion). *Evaluated at x^* , the presumption regime over-deters, meaning that x_P exceeds x^* , when p'/p exceeds $-p'/p$, under-deters when the inequality runs the other way, and implements the optimum at equality.*

The criterion compares the responsiveness of exculpation to care against the responsiveness of harm to care. A rebuttal standard that improves gradually with care yields an elasticity small relative to the hazard elasticity and therefore under-deters, approaching the optimum from below. A rebuttal standard shaped as a sharp threshold, which is what a safe harbour is, yields a large local elasticity and can push care above the social optimum. Section 6.4 develops the design implication.

4.4. Ranking the regimes

The three regimes do not admit an unconditional ranking, and it is worth being explicit about why. Whether the presumption outperforms the claimant's burden depends on both the level and the slope of the rebuttal technology. If rebuttal is easy at every level of care, so that one minus p falls below σ throughout the relevant range, the presumption can deliver weaker deterrence than the claimant's burden despite running the burden in the developer's direction. The comparison that survives without qualification is the one stated in Proposition 3: the burden shift alone is exactly neutral. Everything else depends on the rebuttal standard, which is a design variable rather than a feature of the environment. That is the paper's practical point.

5. Numerical illustration

The table reports equilibrium care and welfare for the baseline parametrisation $k = 1$, $p_0 = 1$, $a = 1$, $H = 100$, which yields $x^* = 3.386$ and a first-best social cost of 9.117. The claimant's probability of proof is set at $\sigma = 0.10$, representing severe verification asymmetry. The care-responsive rebuttal is $p(x) = 0.7x/(x + 2)$; the care-insensitive rebuttal is fixed at $\bar{p} = 0.7$.

Liability regime	Care x	x / x^*	Social cost	Loss vs first best
First best	3.386	1.000	9.117	0.000
Strict liability, claimant's burden	1.746	0.516	18.979	9.862
Negligence, claimant's burden	1.746	0.516	18.979	9.862
Presumption, care-insensitive rebuttal	2.489	0.735	11.396	2.279
Presumption, care-responsive rebuttal	3.037	0.897	9.409	0.292

Three features of the table carry the argument. Negligence and claimant-burden strict liability coincide, because at $\sigma = 0.10$ the negligence threshold of Proposition 2 has been crossed and the standard is

abandoned. The care-insensitive presumption sits at 2.489, which is exactly the level that claimant-burden strict liability delivers at $\sigma = 0.30$, confirming the equivalence of Proposition 3 to the precision of the solver. The care-responsive presumption recovers most of the welfare loss, reducing it from 9.862 to 0.292.

The advantage widens as verification worsens. At $\sigma = 0.30$ the care-responsive presumption reduces the welfare loss by a factor of about eight; at $\sigma = 0.10$ by about thirty-four; at $\sigma = 0.02$ by about one hundred and sixteen. The regime is therefore most valuable precisely where the structural asymmetry is most severe, which is the artificial intelligence case.

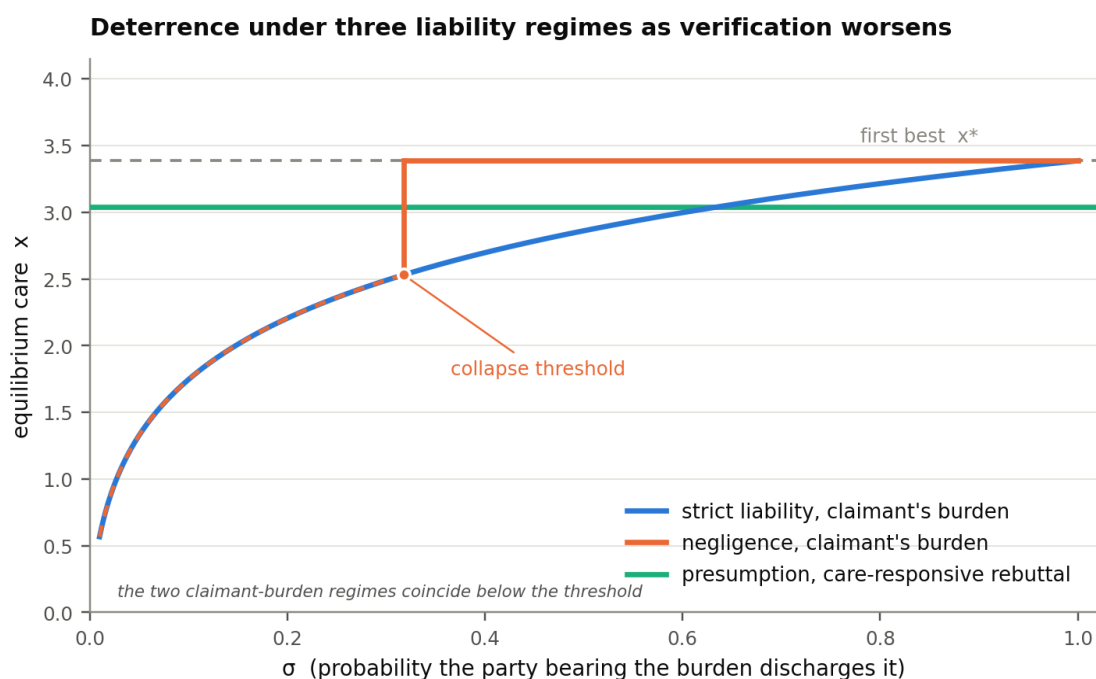


Figure 1. Equilibrium care as the probability of discharging the burden falls. The claimant-burden regimes coincide below the negligence collapse threshold. The presumption line is flat because the developer's incentive under that regime does not depend on the claimant's proof probability.

Figure 1 displays the three regimes across the full range of σ . The negligence path is the one that should give a regulator pause: it is indistinguishable from first best over the upper part of the range and then falls, without warning, onto the claimant-burden path. A rule calibrated on cases where evidence is available carries no information about its performance where evidence is not.

6. Extensions

6.1. Bilateral care

Suppose the affected party also chooses a precaution y at cost $d(y)$, with the probability of harm $q(x, y)$ decreasing in both arguments. The classical objection to strict liability now applies: a fully compensated victim takes no care. The question is whether the presumption regime inherits that defect.

It does not, and the reason is instructive. Under a presumption the victim is not fully compensated in expectation, because the developer rebuts with probability $p(x)$. The victim therefore retains residual exposure equal to $p(x)q(x, y)H$ and has an incentive to take precautions, while the developer retains the exculpatory-evidence incentive of Proposition 4. The following table reports Nash equilibria under $q(x, y) = \exp(-x - y)$ and $d(y) = y^2/2$, with the other primitives as in Section 5.

Regime	Developer x	Victim y	Social cost	Loss vs optimum
Joint optimum	1.965	1.965	5.826	0.000
Strict liability, full ($\sigma = 1$)	3.386	0.000	9.117	3.291
Claimant's burden, $\sigma = 0.3$	1.016	2.370	6.710	0.884
Claimant's burden, $\sigma = 0.1$	0.339	3.047	8.085	2.260
Presumption, care-responsive	2.257	1.186	6.447	0.621

Pure strict liability induces first-best developer care and no victim care at all, and lands further from the joint optimum than either of the diluted regimes. The presumption performs best among the regimes compared, which is a result the baseline unilateral model cannot express. The finding should be read with care: it is specific to the functional forms adopted and to the assumption that the rebuttal probability is the same object that determines the victim's residual exposure. It nonetheless suggests that the presumption's advantage is not confined to the unilateral case.

6.2. Litigation costs and the viability of the claim

Suppose the affected party must incur a filing cost L and sues only when expected recovery covers it. Under the claimant's burden expected recovery is σH ; under the presumption it is $(1 - p(x))H$ evaluated at the equilibrium. With the parameters of Section 5 and $\sigma = 0.30$, expected recovery is 30 under the claimant's burden and approximately 57.79 under the care-responsive presumption.

The consequence is discontinuous. For filing costs between those two values, claims are viable under the presumption and not viable under the claimant's burden. Where claims are not viable, no suit is filed, the developer anticipates this, and equilibrium care falls to zero, with the social cost rising to the full expected harm. The presumption therefore operates on a channel the care analysis alone does not capture: it preserves the existence of the claim, not merely its incentive effect. In a setting where individual harms from a deployed system are small relative to the cost of expert-intensive litigation, this channel may dominate.

6.3. Judgment-proof developers

Suppose the developer's assets A fall short of H , so that it pays the lesser of the two. Both regimes degrade, and the degradation is proportional rather than discontinuous.

Developer assets A	x, claimant's burden	x / x*	x, presumption	x / x*
100 (fully solvent)	2.489	0.735	3.037	0.897
50	2.010	0.594	2.572	0.760
20	1.432	0.423	1.999	0.590
5	0.726	0.214	1.245	0.368

The presumption's advantage survives at every asset level examined but shrinks in absolute terms, and at A = 5 neither regime induces care within reach of the optimum. The implication is that evidentiary design and capital adequacy are complements rather than substitutes: a well-designed rebuttal standard applied to a developer with no assets to reach still fails. For deployment by thinly capitalised entities, the analysis points toward mandatory insurance or channelling liability to a solvent party in the deployment chain, neither of which the present model addresses.

6.4. Endogenous rebuttal and the design of safe harbours

The baseline postulates $p(x)$. A more satisfying treatment derives it from what a producer can document. Suppose the producer's evidence is a signal of x whose informativeness is governed by a sharpness parameter t , and the court exculpates when its posterior that care was adequate exceeds a threshold located at m . A convenient reduced form is

$$\rho(x; m, t) = \rho_{\max} \cdot x^t / (x^t + m^t)$$

As t tends to zero, ρ becomes flat and Proposition 3 applies: the presumption is neutral. As t grows large, ρ approaches a step at $x = m$, which is a bright-line safe harbour: satisfy the specified conditions and exculpation follows. Intermediate t corresponds to a graduated standard under which better documentation improves the producer's position without guaranteeing it.

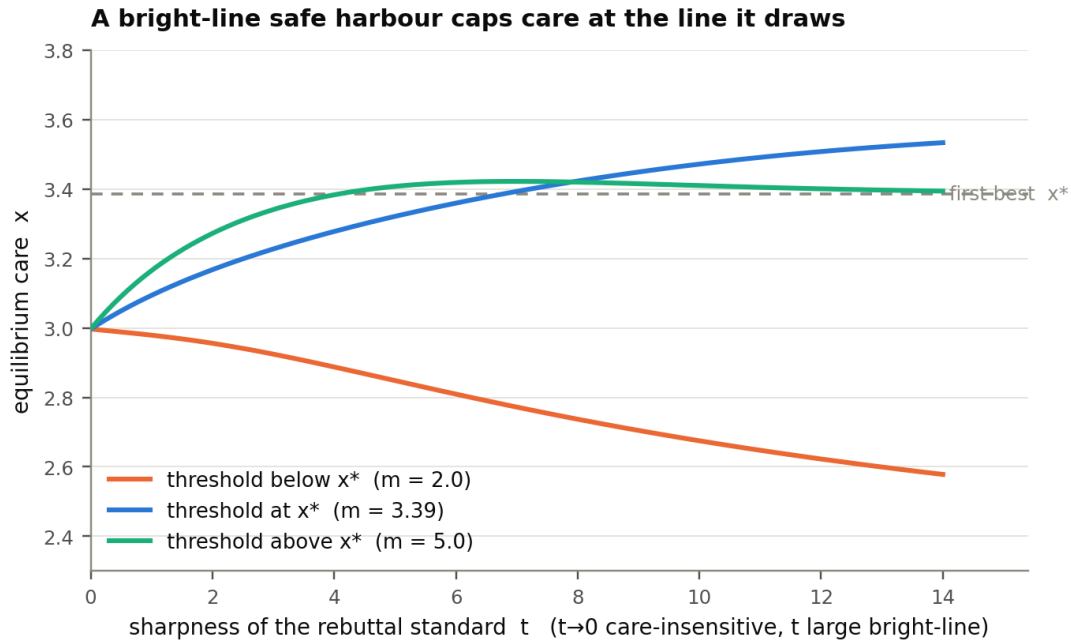


Figure 2. Equilibrium care as the rebuttal standard sharpens, for three locations of the exculpatory threshold. Sharpening helps only when the threshold is located at or above the socially optimal care level.

The design result is visible in the figure. Sharpening the standard is beneficial only when the threshold is located at or above the socially optimal level of care. A bright-line safe harbour set below that level actively destroys deterrence: as the line becomes sharper, equilibrium care converges downward toward the line rather than upward toward the optimum, because the producer's cheapest route to exculpation is to do exactly what the line requires and no more. In the parametrisation shown, a threshold at $m = 2.0$ with high sharpness delivers care of 2.62 against an optimum of 3.386, which is worse than the graduated version of the same standard.

This gives a rule of thumb that does not depend on the particular functional form. A regulator that cannot locate the optimum with confidence should prefer a graduated rebuttal standard to a bright-line one, because a graduated standard errs by approaching the optimum from below while a misplaced bright line caps care at whatever level it happens to specify. The asymmetry of the two errors, rather than the expected accuracy of the threshold, is what should drive the choice.

6.5. Activity levels

Shavell's second margin is the scale at which the injurer operates. Suppose the developer deploys n units, obtaining gross benefit $B(n)$ with B concave, and bearing n times the per-unit care cost and expected harm. Under-internalisation now distorts two decisions rather than one.

Regime	Care x	Scale n	Welfare	Loss vs optimum
Social optimum	3.386	5.581	62.170	0.000
Strict liability, full ($\sigma = 1$)	3.386	5.581	62.170	0.000

Claimant's burden, $\sigma = 0.3$	2.489	9.739	31.454	30.716
Presumption, care-responsive	3.037	7.125	58.656	3.514

A diluted claimant's burden does not merely lower care from 3.386 to 2.489; it raises deployed scale from 5.581 to 9.739, because the developer's private cost per unit understates the social cost. Welfare falls by roughly half. The care-responsive presumption limits the scale distortion as well, holding welfare within six per cent of the optimum. The activity margin therefore roughly doubles the welfare cost of a diluted burden relative to what the care margin alone suggests, which strengthens rather than weakens the case for attending to evidentiary design.

7. Sensitivity

The propositions were re-derived under four parametrisations: the baseline; a polynomial hazard $p(x) = (1 + x)^{-2}$; a cubic cost function $c(x) = x^3/3$; and a slower hazard decay $a = 0.5$. Proposition 3 holds as an exact identity in every case, with the computed difference between the care-insensitive presumption and the corresponding claimant-burden regime equal to zero at solver precision. Proposition 1 holds in every case. The negligence threshold of Proposition 2 exists in every case, falling between 0.236 and 0.318 depending on the parametrisation, so its magnitude is specification-dependent while its existence is not.

Proposition 5 was tested against three rebuttal technologies. A bounded proportional form and an exponential form both under-deter across the parameter ranges examined, consistent with the criterion, since neither generates a rebuttal elasticity exceeding the hazard elasticity at the optimum. A logistic form centred at x^* with slope k over-deters when k exceeds twice the hazard elasticity, and the computed knife-edge coincides with the analytical prediction to four decimal places. Over-deterrence is therefore attainable, but only under a rebuttal standard shaped as a sharp threshold, which is the same feature that Section 6.4 identifies as carrying design risk in the opposite direction.

8. Application to the revised European product liability regime

The European Union withdrew its proposed Artificial Intelligence Liability Directive, a decision confirmed in 2025, leaving the revised Product Liability Directive, Directive (EU) 2024/2853, as the operative instrument for harm caused by software and artificial intelligence. That instrument extends the concept of a product to software and to systems incorporating artificial intelligence, retains liability without fault, and introduces presumptions operating in favour of the claimant where the technical complexity of the case makes proof of defect or causation excessively difficult. Member States have until December 2026 to transpose it.

The architecture matches the third regime of this model. The Directive does not merely restate liability without fault; it addresses the evidentiary margin directly, which is the margin the present analysis identifies as binding. To that extent the instrument is better targeted than the negligence-versus-strict-liability framing that dominates the surrounding commentary.

Proposition 3 nonetheless identifies where the instrument is exposed. The deterrent effect of a presumption depends entirely on whether the rebuttal available to the producer responds to the care actually taken. If a producer can discharge the presumption by producing certification of compliance with applicable regulatory requirements, and if that certification can be obtained without safety investment that varies with the risk of the deployed system, then p becomes care-insensitive, p' falls to zero, and the regime collapses into diluted liability without fault. The presumption would then redistribute litigation risk while leaving the incentive to invest in safety where it stood. Section 6.4 adds a second exposure: even a care-responsive rebuttal standard will cap care at the level it specifies if it is drafted as a bright line and that line is set below the level of care the risk warrants.

This yields two falsable predictions about transposition. Where Member States implement the rebuttal so as to require the producer to adduce evidence of system-specific safety work, such as documented evaluation against the failure modes at issue, the model predicts care converging toward the social optimum. Where transposition permits rebuttal by formal certification alone, the model predicts equilibrium care indistinguishable from a regime in which the claimant had retained the burden throughout. The two predictions differ observably in producer safety expenditure and in the disposition of contested cases. Divergence in transposition across Member States around the December 2026 deadline offers a setting in which to test them.

9. Policy implications

Three implications follow for the design of artificial intelligence liability rules, stated in decreasing order of confidence.

The first is that the direction of the burden is the wrong object of debate. Shifting it onto the developer is neither sufficient nor, by itself, useful. What matters is the content of the rebuttal standard, and legislative attention spent on the former at the expense of the latter is misdirected. This follows from Proposition 3, which is an identity and does not depend on parametrisation.

The second is that a graduated rebuttal standard is preferable to a bright-line safe harbour under uncertainty about the socially optimal level of care. The two err asymmetrically: the graduated standard approaches the optimum from below, while a misplaced bright line caps care at the line. This follows from Section 6.4 and depends on the functional form only through the direction of the effect, not its magnitude.

The third is that evidentiary design does not substitute for solvency requirements. The presumption's advantage persists as developer assets fall but shrinks, and against a sufficiently thinly capitalised developer no evidentiary rule induces adequate care. Where systems are deployed by entities whose assets are small relative to the harm they can cause, the analysis points toward insurance or liability channelling, which lie outside the present model.

10. Limitations

The model treats litigation as a single probabilistic event rather than as a process with settlement, discovery, and appeal, each of which could alter the effective proof probabilities in ways the reduced form suppresses. The rebuttal technology of Section 6.4 is a reduced form rather than a derivation from an explicit Bayesian court, and constructing that court is the natural next step. Insurance is absent, and since insurance both relaxes the judgment-proof constraint and dulls the care incentive, its inclusion could change the ranking in Section 6.3. The bilateral-care result of Section 6.1 is specific to the functional forms adopted and to the identification of the victim's residual exposure with the developer's rebuttal probability, and it should be treated as suggestive rather than established. Multiple potential injurers along a deployment chain, which is the ordinary situation for a model built by one firm, fine-tuned by another, and deployed by a third, are not modelled at all.

The characterisation of Directive (EU) 2024/2853 in Section 8 operates at the level of the instrument's architecture as reported in the comparative literature. The argument there concerns the structure of a rebuttable presumption in general; any claim about the operation of specific provisions would require analysis of the transposing national measures, which do not yet exist in final form.

11. Conclusion

Classical accident theory compares liability rules on the care margin because that is where its informational friction sits. For harm caused by artificial intelligence the binding friction sits elsewhere, on the question of who can prove what, and the comparison changes accordingly. A claimant's burden dilutes deterrence smoothly toward zero. A negligence standard fails discontinuously and demands more verification than the alternative it is usually compared against. Shifting the burden onto the developer appears to solve the problem and, by itself, does not: a presumption rebuttable on grounds unrelated to care reproduces the regime it replaced, exactly.

What restores deterrence is a rebuttal standard under which care buys evidence, so that the developer's investment in safety and its capacity to exculpate itself move together. The extensions suggest that this property does more than fix the care margin: it preserves the viability of claims that filing costs would otherwise extinguish, it limits the scale distortion that under-internalisation produces, and it leaves the victim enough residual exposure to take precautions of their own.

The design question for artificial intelligence liability is therefore narrower and more tractable than the debate suggests. It is not whether to adopt negligence or liability without fault. It is what a producer must be required to show in order to escape a presumption, and the answer that preserves deterrence is: evidence of the safety work that the presumption exists to encourage, presented against a graduated standard rather than a bright line.

References

Calabresi, G., & Melamed, A. D. (1972). Property rules, liability rules, and inalienability: One view of the cathedral. *Harvard Law Review*, 85(6), 1089–1128.

- Coase, R. H. (1960). The problem of social cost. *Journal of Law and Economics*, 3, 1–44.
- Dawkins, R. (1982). *The extended phenotype*. Oxford University Press.
- Demougin, D., & Fluet, C. (2006). Preponderance of evidence. *European Economic Review*, 50(4), 963–976.
- Directive (EU) 2024/2853 of the European Parliament and of the Council on liability for defective products, repealing Directive 85/374/EEC. *Official Journal of the European Union*.
- Guerra, A., Luppi, B., & Parisi, F. (2022). Do presumptions of negligence incentivize optimal precautions? *European Journal of Law and Economics*, 54, 349–368.
- Hay, B. L., & Spier, K. E. (1997). Burdens of proof in civil litigation: An economic perspective. *Journal of Legal Studies*, 26(2).
- Hull, D. L. (1988). *Science as a process*. University of Chicago Press.
- Kaplow, L. (2011). On the optimal burden of proof. *Journal of Political Economy*, 119(6), 1104–1140.
- Lakatos, I. (1978). *The methodology of scientific research programmes*. Cambridge University Press.
- Lerer, I. A. (2025a). Law as extended phenotype: Toward an evolutionary theory of legal systems. SSRN.
- Lerer, I. A. (2025b). Formal foundations of multilevel game theory. SSRN 5974555.
- Lerer, I. A. (2026a). Transaction costs, extended phenotype, and liability for artificial intelligence: A common conceptual framework. Zenodo. <https://doi.org/10.5281/zenodo.21562785>
- Popper, K. R. (1959). *The logic of scientific discovery*. Routledge.
- Shavell, S. (1987). *Economic analysis of accident law*. Harvard University Press.

AI tools were used as research support during the preparation of this paper: locating sources, extracting information from the existing literature, verifying the numerical solution of the model, and reviewing drafts. The thesis, analytical framework, arguments, and conclusions constitute the original contribution of the author. This is an independent research project; all intellectual responsibility for the content rests with the author.

Appendix A. Proofs

Throughout, c is increasing and strictly convex with $c(0) = 0$ and $c'(0) = 0$; p is decreasing and strictly convex; H is strictly positive.

Proof of Proposition 1. Let $V(x) = c(x) + \sigma p(x)H$. Then $V''(x) = c''(x) + \sigma p''(x)H$, which is strictly positive for σ at least zero, so V is strictly convex and its minimiser x_C is unique and characterised by $c'(x_C) = -\sigma p'(x_C)H$. Write $F(x, \sigma) = c'(x) + \sigma p'(x)H$. Then $\partial F/\partial x = c'' + \sigma p''H$ is strictly positive and $\partial F/\partial \sigma = p'H$ is strictly negative, so by the implicit function theorem $dx_C/d\sigma = -(\partial F/\partial \sigma)/(\partial F/\partial x)$ is strictly positive. At $\sigma = 1$ the condition coincides with that defining x^* , so x_C equals x^* ; by monotonicity x_C is strictly less than x^* for σ less than one. As σ tends to zero the condition becomes $c'(x) = 0$, whose solution is $x = 0$ given $c'(0) = 0$ and c' strictly increasing. ■

Proof of Proposition 2. Compliance yields payoff $c(\bar{x})$. Define $S(\sigma_N)$ as the minimum of $c(x) + \sigma_N p(x)H$ over x in the interval from zero to $\bar{x}$, excluding $\bar{x}$ itself. By the envelope theorem $dS/d\sigma_N = p(x_s)H$, strictly positive, so S is continuous and strictly increasing, with $S(0) = c(0) = 0$. Compliance is optimal exactly when $c(\bar{x})$ is no greater than $S(\sigma_N)$. At $\sigma_N = 0$ this fails, since $c(\bar{x})$ is strictly positive. At $\sigma_N = 1$, $S(1)$ equals the infimum of $c(x) + p(x)H$ on that interval, which exceeds $c(\bar{x})$ whenever $\bar{x}$ is set at or below x^* , so the condition holds. Continuity and strict monotonicity give a unique $\hat{\sigma}$ in the open unit interval at which the two sides are equal, with compliance above and shirking below. When shirking, the developer solves the problem of Proposition 1 with σ replaced by σ_N , and the constraint x below $\bar{x}$ does not bind because the unconstrained solution is already below $\bar{x}$ in that region. For the ordering of proof probabilities, note that the event in which the claimant establishes breach requires establishing defect and causation together with a shortfall against the standard, so it is contained in the event in which the claimant establishes defect and causation alone; hence σ_N is no greater than σ . ■

Proof of Proposition 3. With $p(x)$ equal to a constant $\bar{p}$, the developer's objective becomes $c(x) + (1 - \bar{p})p(x)H$, whose first-order condition is $c'(x) = -(1 - \bar{p})p'(x)H$. This is the first-order condition of Proposition 1 with $\sigma = 1 - \bar{p}$. Both objectives are strictly convex in x , so each has a unique minimiser and the two minimisers coincide. The equivalence is exact for every admissible $\bar{p}$. ■

Proof of Proposition 4. Let $W(x) = c(x) + p(x)(1 - p(x))H$. Then $W'(x) = c'(x) + [p'(x)(1 - p(x)) - p(x)p'(x)]H$, so the first-order condition is as stated in the text. Fix a point x_0 and compare with the care-insensitive regime holding p fixed at $p(x_0)$. The marginal private benefit of care differs by exactly $p(x_0)p'(x_0)H$, which is strictly positive when p' is strictly positive. Since c' is strictly increasing and, by the assumed second-order condition, W is strictly convex on the relevant range, the minimiser under the care-responsive rebuttal lies strictly above that under the care-insensitive one. ■

Proof of Proposition 5. Write the marginal private benefit of care under the presumption as $MB_P(x) = -p'(x)(1 - p(x))H + p(x)p'(x)H$, and the marginal social benefit as $MB^*(x) = -p'(x)H$. Their difference is $MB_P(x) - MB^*(x) = H[p'(x)p(x) + p(x)p'(x)]$, which equals H times the derivative of the product $p(x)p(x)$. Evaluating W' at x^* , and using $c'(x^*) = MB^*(x^*)$, gives $W'(x^*) = MB^*(x^*) - MB_P(x^*) = -H \cdot (pp)'(x^*)$. If $(pp)'(x^*)$ is strictly positive then $W'(x^*)$ is strictly negative and the minimiser lies strictly above x^* , which is over-deterrence; if strictly negative, the minimiser lies below x^* . The sign condition $(pp)'$ greater than zero is equivalent to $p'p + pp'$ greater than zero, hence to pp' greater than $-p'p$, hence, dividing by the strictly positive product pp , to p'/p greater than $-p'/p$. ■

Appendix B. Computational note

Equilibria were obtained by solving each first-order condition with Brent's method over the interval from zero to sixty, with a bounded-minimisation fallback where the derivative does not change sign on that interval. Bilateral-care equilibria were computed by iterated best response to a tolerance of $1e-10$; activity-level equilibria by bounded numerical minimisation over the pair consisting of care and scale. The social optimum and each regime were solved under four parametrisations of the cost and hazard functions and three rebuttal technologies. The equivalence asserted in Proposition 3 was verified as an exact identity, with the absolute difference between the two computed equilibria equal to zero at machine precision in every parametrisation tested. The knife-edge of Proposition 5 was located numerically at the point predicted analytically. The scripts that generate every number and figure reported here accompany this paper; they depend only on NumPy, SciPy, and Matplotlib.